MOST IMPORTANT C PROGRAMMING QUESTIONS
POKHARA UNIVERSITY
BCA 2nd SEMESTER

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Last edited *on Aug 13 2025*

Theory Questions

1. What are the key features that make C a popular and widely used programming language? Explain the basic structure of a C program with an example. **(Repeated 2 times)**
2. Differentiate between single line and block comments in C with an example. Why are comments important in a program? Explain.
3. What is a looping statement? Explain while loop and do-while loop using a suitable example. **(Repeated 2 times)**
4. Define a switch statement with syntax. Write a program to display a pattern.
5. What is an array? Explain with examples. **(Repeated 3 times)**
6. Define String. Explain six string handling functions with examples. **(Repeated 2 times)**
7. What are the advantages of using functions? Discuss local and global variables with examples. (**Repeated 3 times)**
8. What are pointers? Explain Dynamic memory allocation and deallocation using pointers. **(Repeated 2 times)**
9. Why is file handling necessary in a C program? Explain various file opening modes in C.
10. Define a multidimensional array. How is it declared and initialized? **(Repeated 3 times)**
11. What is a recursive function? Explain with an example.
12. What are variables and keywords? Give examples. **(Repeated 4 times)**
13. List out the different types of operators. Illustrate bitwise operators with examples. **(Repeated 4 times)**
14. What is Top-Down Analysis? Why is decomposition of a project into different modules necessary in programming? **(Repeated 2 times)**
15. What is a data flow diagram? Explain the importance of decomposing a project into various modules.
16. Why is C called a middle-level language? Explain C program execution with a suitable diagram. **(Repeated 2 times)**
17. What is a datatype? List out data-types available in C. Explain assignment and ternary operators with examples. **(Repeated 2 times)**
18. What is a function? List its components.
19. What is a loop? Explain with examples.
20. What are character arrays? How are they different from integer arrays? Explain 5 string functions with examples.
21. What is call by reference? Explain with examples. **(Repeated 2 times)**
22. Explain the continue and break statements. **(Repeated 2 times)**
23. What is a nested loop? Explain with examples.
24. Why do we declare a variable? List the rules for naming variables. Provide data types and declare variables for specific cases.
25. What is operator overloading? List out operators available in C. Explain increment and decrement operators with examples. **(Repeated 2 times)**

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Programming Questions

1. Write a program to calculate prime numbers between 1 to 50. **(Repeated 3 times)**
2. Write a program to display the following pattern:

```
*
**
***
****
*****
```

3. Write a program to input any 10 numbers and find the maximum and minimum among the**m. (Repeated 2 times)**
4. Write a program to get 20 numbers from the user and sort them in ascending order. **(Repeated 2 times)**
5. Write a program to generate a Fibonacci series of \(n \) numbers using a recursive function**. (Repeated 4 times)**
6. Write a program to open a file named "teachers.txt" to keep records of faculty members (eid, name, address, department, mobile) and insert records into the file. Display all records of teachers on the screen. **(Repeated 2 times)**
7. Write a program to input name, address, roll, and age of 100 students and display the information of students whose age is less than or equal to 18 using structure. **(Repeated 3 times)**
8. Write a program to input elements of a 3x3 matrix and find the sum of diagonal elements using arrays and pointers.
9. Write a program to search a given element in an array consisting of \(N \) elements.
10. Write a program to read a line of text and print each character in reverse case (lowercase if uppercase and vice versa). **(Repeated 2 times)**
11. Write a program to display the following pattern:

```
P O K H A R A
  O K H A R
    K H A
      H
```

Or

```
P O K H A R A
O K H A R
K H A
H
```

Or

```
P O K H A R A
      O K H A R
            K H A
                  H
```

12. Write a menu-driven program to perform the following tasks:
    - Check if a given number is even or odd.
    - Check if a given number is prime or composite.
    - Check if a given number is positive or negative.
    - Exit the program.
13. Write a program to find the factorial of a given number using a user-defined function.
14. Write a program to find the smallest and largest number in an array.
15. Write a program to transpose a 3x3 matrix. **(Repeated 4 times)**

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